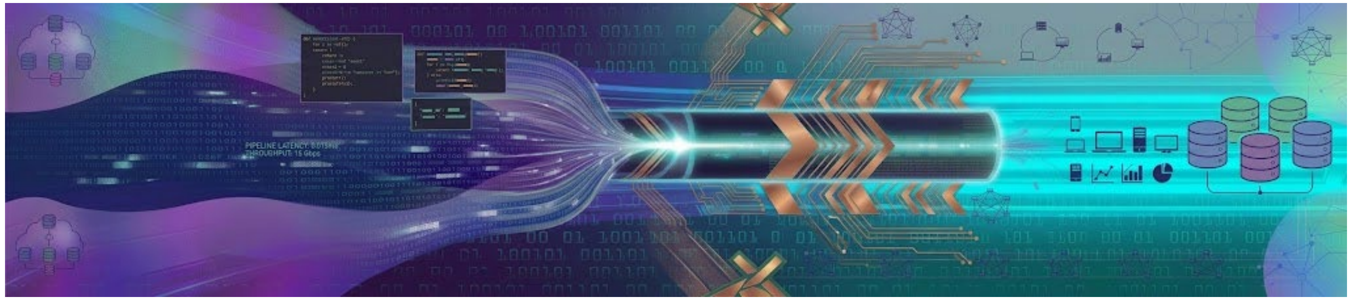
High Probability Streaming Lower Bounds for F_2 Estimation



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Streaming Model

- Input: Elements of an underlying data set S, updates arrive sequentially

- Output: Evaluation (or approximation) of a given function

Internet traffic

Sensor networks

Stock markets

Frequency Vector

- Given a set S of m elements from $[n]$, let f_i be the frequency of element i. (How often it appears)

$$1\ 1\ 2\ 1\ 2\ 1\ 1\ 2\ 3 \rightarrow [5, 3, 1, 0] := f$$

- Insertion-only stream: each update only increases coordinates

- Goal: Use space *sublinear* in the stream length m and dimension n

Frequency Moments

- Given a set S of m elements from $[n]$, let x_i be the frequency of element i. (How often it appears)
- Let F_2 be the frequency moment of the vector:

$$F_2 = |x_1|^2 + |x_2|^2 + \ \ldots + |x_n|^2$$

- Goal: Given a set S of m elements from $[n]$ and an accuracy parameter ε, output a $(1+\varepsilon)$-approximation to F_2
- Motivation: Entropy estimation, linear regression, subroutine for many other problems

F_2 Estimation

- AMS algorithm gives $(1+\varepsilon)$-approximation to F_2 using $O\left(\frac{1}{\varepsilon^2}\log n\right)$ bits of communication [AlonMatiasSzegedy99]

- Let $s \in \{-1,+1\}^n$ be a random sign vector
- Let $Z = \langle s, f\rangle = s_1 f_1 + s_2 f_2 + \cdots + s_n f_n$

- $Z^2 = \sum_{i,j\in[n]} s_i s_j f_i f_j$
- $\mathbb{E}[Z^2] = (f_1^2 + f_2^2 + \cdots + f_n^2) := F_2$

AMS Algorithm

- $\mathrm{E}[Z^2] = \sum_{i,j\in[n]} s_i s_j f_i f_j$
- $\mathrm{Var}[Z^2] \leq \mathrm{E}[Z^4] = \sum_{a,b,c,d\in[n]} s_a s_b s_c s_d f_a f_b f_c f_d \leq 6F_2^2$

- Chebyshev's inequality: the average of $O\left(\frac{1}{\varepsilon^2}\right)$ independent instances gets $(1+\varepsilon)$-approximation with probability at least $\frac{2}{3}$

- Chernoff bounds / median-of-means: $(1+\varepsilon)$-approximation to F_2 moment with probability $1-\delta$, using $O\left(\frac{1}{\varepsilon^2}\log n \log\frac{1}{\delta}\right)$ bits of communication [AlonMatiasSzegedy99]

Lower Bounds

- $\Omega\left(\frac{1}{\varepsilon^2} + \log n\right)$ lower bound from Gap-Hamming and INDEX [Woodruff04]

- $\Omega\left(\frac{1}{\varepsilon^2} \log n\right)$ lower bound from Exam-Set-Disjointness [BravermanZamir26]

Our Results

- $\Omega\left(\frac{1}{\varepsilon^2}\log n \log\frac{1}{\delta}\right)$ lower bound
  - New communication problem called Exam-Mostly-Set-Disjointness
  - $\Omega\left(\frac{m}{t}\right)$ lower bound for Exam-Mostly-Set-Disjointness

- Better algorithms for bounded support and bounded frequency
- $\tilde{O}\left(\frac{1}{\varepsilon^2}\log^3\frac{B}{\varepsilon}\log\frac{1}{\delta}\right) + \mathrm{polylog}(n)$ space when all coordinates at most B
- $\tilde{O}\left(\frac{1}{\varepsilon^2}\log\frac{k}{\varepsilon}\log\frac{1}{\delta} + \log n\right)$ space when support at most k

Communication Complexity

- Multi-party, one-way model: t players $P_1, \ldots, P_t$ get inputs $X_1, \ldots, X_t$ drawn from a joint distribution and must compute $f(X_1, \ldots, X_t)$
- First player P_1 considers input X_1 and sends a message M_1 to P_2
- Each player P_i considers M_{i-1} and X_i and sends a message M_i to P_{i+1}

- The protocol Π has a transcript $\Pi(X)$ and the communication cost is the length of $\Pi(x)$, i.e., the total communication across all players

(Distributional) Set Disjointness

- Universe of size n, number of players t
- YES case:
  - For each item in $[n]$:
    - With probability $\frac{1}{2}$, assign it to no one
    - With probability $\frac{1}{2}$, assign to a random player in $[t]$
- NO case:
  - For a special random item, assign it to ALL players
  - For each other item in $[n]$:
    - With probability $\frac{1}{2}$, assign it to no one
    - With probability $\frac{1}{2}$, assign to a random player in $[t]$

Complexity of Set Disjointness

- In the NO case, there is a common item among ALL players, and for sufficiently large universe size n, each player has a set of size $\Theta\left(\frac{n}{t}\right)$
- First player can communicate all elements using $O\left(\frac{n}{t}\log n\right)$ communication
- Better approaches where players iteratively remove candidates for the common intersection [HastadWidgerson07] using total communication $O\left(\frac{n}{t}\right)$
- Lower bound $\Omega\left(\frac{n}{t}\right)$ [ChakrabartiKhotSun03]

Set Disjointness and F_2

- $O\left(\frac{n}{t}\right)$ total communication across t players, so it could be possible that each player sends $O\left(\frac{n}{t^2}\right)$ bits
- Can be simulated by a streaming algorithm with space $O\left(\frac{n}{t^2}\right)$

- YES case, $\Theta(n)$ coordinates each appear once → $\mathrm{F}_2 = \Theta(n)$
- NO case, also a single coordinate that appears t times
- $n + t^2 \approx (1+\varepsilon)n$, so $t^2 = \varepsilon n$
- Only gives lower bound of $\Omega\left(\frac{1}{\varepsilon}\right)$

Exam Set Disjointness [BravermanZamir26]

- Universe of size n, number of players $t+1$, last player is a referee
- Inputs to all other players the same
- Referee has special item $i \in [n]$ and must decide whether set intersection is disjoint or equal to i

- Harder communication problem, so still lower bound $\Omega\left(\frac{n}{t}\right)$

Boosting the Gap [BravermanZamir26]

- Special item i used to "boost" the gap between YES/NO cases
- Referee adds $\frac{t}{\varepsilon}$ copies of i
- YES case, $\Theta(n)$ coordinates each appear once, single item appears $\frac{t}{\varepsilon}$ times → $\mathrm{F}_2 = \Theta\left(n + \frac{t^2}{\varepsilon^2}\right) = \Theta(n)$ when $t^2 = \varepsilon^2 n$
- NO case, also same coordinate appears t times
- $\mathrm{F}_2 = \Theta\left(n + \left(t + \frac{t}{\varepsilon}\right)^2\right) \approx (1+\varepsilon)n$, so distinguishable
- Lower bound of $\Omega\left(\frac{1}{\varepsilon^2}\right)$

Packing Multiple Instances

- Have a number of different instances, with $t = 2,4,8, \ldots, O(\log n)$ players
- Lower bound of $\Omega\left(\frac{1}{\varepsilon^2}\right)$ for each instance, independent of t
- Direct sum argument across instances gives $\Omega\left(\frac{1}{\varepsilon^2}\log n\right)$ lower bound

- A few caveats: instances are not independent, use tuples of elements instead, instances are "interweaved" in the data stream
- Overall good template, just need "harder" communication problem

Mostly Set Disjointness

- Universe of size n, number of players t
- YES case, M-almost disjoint:
  - At most M repeated items across all players
- NO case:
  - For a special random item, assign it to $\Theta(t)$ players
  - All other items, at most M repeated items across all players

Mostly Set Disjointness

- Universe of size n, number of players t
- YES case, M-almost disjoint:
  - For each item in $[n]$:
    - For each player in $[t]$, assign the item with probability $\Theta\left(\frac{1}{t}\right)$
- NO case:
  - For a special random item, assign it to $\Theta(t)$ players
  - For each other item in $[n]$:
    - For each player in $[t]$, assign the item with probability $\Theta\left(\frac{1}{t}\right)$
- $M = O\left(\log \frac{1}{\delta}\right)$

Exam Mostly Set Disjointness

- Mostly set disjointness problem, but also extra player (referee)
- Inputs to all other players the same
- Referee has special item $i \in [n]$ and must decide whether mostly set intersection is disjoint or equal to i

- We show this problem has lower bound $\Omega\left(\frac{n}{t}\log\frac{1}{\delta}\right)$

Exam Mostly Set Disjointness

- One bit problem $\mathrm{F}_{ct,t}$, each player receives a single bit $\{0,1\}$
- YES case, Hamming weight at most 1
- NO case, Hamming weight at least ct

- Suffices to measure conditional information cost on YES case
  - With probability $\frac{1}{2}$, input is 0^t
  - With probability $\frac{1}{2}$, input is e_i, when auxiliary random variable $R = i$

- $\Omega(1)$ known for $\delta = 2^{-\Theta(t)}$, $\Omega\left(\frac{1}{t}\log\frac{1}{\delta}\right)$ by amplification argument

Hard Distribution to Product Distribution

- Define μ_p as the distribution where each coordinate is set to 1 with probability $p = \Theta\left(\frac{1}{t}\right)$

- Constant probability of all zeros vector, $(1-p)^t$
- Probability $\Theta(p)$ of any specific elementary vector e_i, so probability $\Theta(pt)$ of seeing *some* elementary vector

- Coupling argument to show lower bound

$$\Theta(pt) \cdot \Omega\left(\frac{1}{t}\log\frac{1}{\delta}\right) = \Omega\left(p\log\frac{1}{\delta}\right) = \Omega\left(\frac{1}{t}\log\frac{1}{\delta}\right)$$

Direct Sum

- Universe size n
- Single coordinate problem has information cost $\Omega\left(\frac{1}{t}\log\frac{1}{\delta}\right)$
- Overall information cost $\Omega\left(\frac{n}{t}\log\frac{1}{\delta}\right)$

Summary and Next Directions

- $\Omega\left(\frac{1}{\varepsilon^2}\log n\log\frac{1}{\delta}\right)$ lower bound
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- Better algorithms for bounded support and bounded frequency
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- Question: High probability bounds for all frequency moments F_p?